

# Samonwita Sarker

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## SUMMARY

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- I'm a CSE undergraduate at CUET, graduating in 2027, who builds full-stack web applications with React, Node.js, and PostgreSQL. I recently worked as an AI evaluation contractor at Fleet AI, Inc., authoring and QA-reviewing long-horizon agentic AI evaluation tasks. Along the way, I've published two Chrome extensions on the Web Store, co-authored an IEEE paper on explainable AI, and written a first-author system paper that placed 7th in a shared task at ACL 2026. **Open to software engineering and ML job opportunities: remote, or on site after graduating in 2027.**

## EDUCATION

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- **B.Sc. in Computer Science and Engineering** (2023 – 2027, expected)  
Chittagong University of Engineering and Technology (CUET) — **CGPA: 3.51/4.00, rising: latest term 3.85**
- **Higher Secondary Certificate (HSC)** (2021)  
Viqarunnisa Noon College — **GPA: 5.00/5.00**

## EXPERIENCE

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- **Team Lead, Crawler Team (Industrial Attachment)** (3 Sep 2026 – Present)  
BdREN Innovation - Cortex — [\[GitHub\]](#)  
Cortex crawls a website, embeds it into a vector database, and answers questions about it with citations. **Lead the five-person crawler team**, splitting the work into portions with named owners and taking it through review. Built its **CUET scraper in Python**: the site is a Next.js app whose pages ship no content, so it reads the public JSON API first and falls back to a headless browser, capturing **735 documents, 382 faculty profiles, and 1,106 linked files** with source provenance for every citation and a 26-check verification stage. Added a per-host rate limiter, skip and failure records, and 13 tests to the team's shared crawler, and wrote the data guide the knowledge team builds on.
- **AI Evaluation Contractor (Generalist)** (May 2026 – Aug 2026)  
Fleet AI, Inc. (Remote)  
Authored long-horizon, multi-step agentic AI evaluation tasks across multiple simulated enterprise application environments, grounding realistic workflows in verified environment data. Conducted QA review on other contributors' tasks, assessing must-have/should-have coverage, action counts, ambiguity, codename compliance, and project-specific rules to ensure correctness, novelty, and robust edge-case coverage.
- **Web Developer (Freelance)** (2025)  
PIERRA, Montreal (Remote) — [\[Website\]](#)  
Designed and shipped a bilingual (English/French) business website for a Montreal-based exterior design firm using **Next.js and Tailwind CSS**, from brief to deployment. Kept all copy in one locale-keyed layer so the two languages stay in step, and built an interactive project gallery, a testimonial carousel, and a consultation request form.

## SKILLS

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- **Languages:** C++, JavaScript (ES6+), TypeScript, Python, Java, C, SQL.
- **Web Development:** React.js, Node.js, Next.js, Express.js, PostgreSQL, Firebase, Tailwind CSS, Three.js, shadcn/ui.
- **AI & Machine Learning:** PyTorch, scikit-learn, Transformers, NLP, Multimodal Deep Learning, Explainable AI (SHAP), Gemini API.

- **Systems & Architecture:** MVC Architecture, Data Structures, Algorithms, Object-Oriented Programming (OOP), DBMS, OS Scheduling.
- **Tools:** Docker, Git/GitHub, TanStack Router, LaTeX, Linux.

## PROJECTS

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- **RIPHours: Productivity Chrome Extension** [\[Chrome Web Store\]](#) — [\[GitHub\]](#)  
**Solo Developer.** Built and published a privacy-first Chrome Extension using **JavaScript, Chrome APIs, and Chrome Storage Sync**. Engineered accurate time-tracking via the Page Visibility API with auto-idle detection, per-site daily limits with hard-block enforcement, and a 7-day trend analytics dashboard. Implemented zero-network-request architecture ensuring all user data remains local.
- **Greenlight: Browser-Driven Robot Arm Simulator** [\[Simulator\]](#) — [\[GitHub\]](#)  
**Lead Developer, IUT Techathon 2026 (Finalist).** Built a 3D robot arm simulator in **Next.js, Three.js, TypeScript, and Zustand** where six input methods (joystick, keyboard, voice keyword, a Gemini voice agent, autonomous PIN entry, and natural language) all resolve to one motion command and run through a single pipeline: an **inverse-kinematics planner**, a **safety gate** that validates every move, and an executor. Wrote the kinematics, the safety gate, the planner and executor, and every input path.
- **TabSaver: Session Management Chrome Extension** [\[Chrome Web Store\]](#) — [\[GitHub\]](#)  
**Solo Developer.** Developed and published a browser productivity extension using **JavaScript and Chrome Extension APIs**. Implemented session persistence and one-click restoration that lets users close entire tab groups and reopen them later, freeing the memory held by open tabs without losing context.
- **ResumeForge: Dynamic Resume Builder** [\[Resume Builder\]](#) — [\[GitHub\]](#)  
**Solo Developer.** Built a complete resume-building web application using **React, Tailwind CSS, and React Router**. Engineered a real-time live-preview engine with multiple professional templates, client-side PDF export, and auto-save via local storage using React Context API.
- **Sixpence: Personal Finance Ledger** [\[Live App\]](#) — [\[GitHub\]](#)  
**Solo Developer.** Built and deployed a personal finance application in **PHP 8 and PostgreSQL** with no framework and no third-party dependencies, writing the PDF export engine and SMTP mail client from scratch. Engineered pace-aware budgets and like-for-like month-to-date comparisons, alongside recurring income and bills, savings goals, and CSV/PDF reports across 12 display currencies. Hardened it with prepared statements throughout, per-session CSRF tokens, account lockout after 5 failed sign-ins, and a strict Content Security Policy, all covered by a **68-test suite** running on PHP 8.1–8.4 against PostgreSQL 16 in CI.
- **DimSum: LLM-Assisted Campus Energy Optimizer** [\[Live Console\]](#) — [\[GitHub\]](#)  
**Interpreter and Guardrails, BUP CSE Fest 2026 Preliminary (Team PTSD).** Built stages two and three of a six-stage **Bun and TypeScript** service that turns plain-English campus operator notes into a provably legal 24-hour energy plan: the **Gemini** call converting each note into a structured directive, and the deterministic guardrail layer that repairs or discards unsafe model output, with a retry ladder and regex fallback for rate limits. Cut median model latency from 5.7s to 2.3s. All ten published scenarios reproduce the organisers' reference cost exactly.
- **MediHub: AI-Powered Hospital Management** [\[Healthcare Portal\]](#) — [\[GitHub\]](#)  
**Full-Stack Developer.** Engineered a healthcare management platform using **React, Node.js, and Firebase**. Integrated the **Gemini API** to summarize uploaded medical reports into key findings, test results, and recommendations, so patients and doctors read a condensed summary instead of the full record. Implemented role-based access control (RBAC) across patient, doctor, and admin dashboards to enforce per-role data privacy.
- **AnomLite: Lightweight Crime Detection in Surveillance Video** [\[GitHub\]](#)  
**Solo Developer.** Built a hybrid **MobileNetV2 + LSTM** model in **PyTorch** for 14-class crime detection on the UCF-Crime dataset, kept to roughly 11 million parameters so it suits real-time use on constrained hardware. Reached **79% accuracy, 0.97 ROC AUC, and 0.75 macro F1**.

- **StockMaster: Enterprise Inventory Tracker** [\[GitHub\]](#)  
**Full-Stack Developer.** Built an inventory management web application using **Next.js**, **TypeScript**, and **Tailwind CSS**. Engineered a reusable component architecture with custom hooks and centralized state management for real-time stock auditing and data visualization.
- **Adaptive Priority Round Robin Simulator** [\[GitHub\]](#)  
**Systems Programmer.** Developed a custom OS scheduling algorithm simulation in **C++** to measure CPU performance metrics such as turnaround and average wait time. Designed a **React** frontend that animates the dynamic process queue and plots these metrics live as scheduling parameters change.

## RESEARCH & PUBLICATIONS

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- **A Machine Learning and Explainable AI-Based Multiclass Police Fraud Prediction Scheme with SHAP Based Interpretability**  
M. Chowdhury, J. Islam, Md. A. I. Semon, **S. Sarker**, M. M. Barua, A. Akter  
2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE), IEEE  
DOI: <https://ieeexplore.ieee.org/document/11429440>
- **7th Rank, DravidianLangTech @ ACL 2026 Shared Task: Telugu Prompt-Style Recovery** (Feb 2026)  
**S. Sarker**, I. M. Sami, P. Mojumder, A. Mallik, H. Murad  
First-author system paper. Classified Telugu text into **nine communicative styles**, comparing four multilingual transformers trained with focal loss; the **MuRIL** system placed 7th with a macro F1 of 0.1703, against a chance level of about 0.11. Published in the Proceedings of DravidianLangTech @ ACL 2026. [\[Paper\]](#) — [\[Code\]](#)
- **Finalist, PoliMemeDecode Datathon (CUET CSE FEST 2025)** (Dec 2025)  
Built a multimodal classifier for political memes, fusing visual and textual features (DenseNet-121 and BanglaBERT, with EasyOCR) to reach a macro F1 of 0.9008.

## COMPETITIONS

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- **Finalist**, IUT Techathon 2026 — built Greenlight, above.
- **Finalist**, AI Hackathon, SciBlitz 2.0 (IEEE CUET Student Branch with SheSTEM), 2026.
- **Finalist**, The Infinity AI BuildFest 2026.
- **Finalist**, MicroOps Hackathon 2025 — a full-stack MVP, API and client, built inside 24 hours.

## CERTIFICATIONS

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- **All Certifications** [\[View All on Google Drive\]](#)  
Full-Stack Development (freeCodeCamp), Data Science & Python (DataCamp), Responsive Web Design, JavaScript Algorithms and Data Structures.

## LEADERSHIP & EXTRACURRICULAR

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- **President, CUET MUN Club** (Present)  
Leads the club and its national conferences. Represented delegations in committee sessions, focusing on international technological governance and resolution drafting.
- **Vice President (Organizing), CUET Computer Club (CCC)** (2023 – Present)  
Competed in inter-university programming contests and collaborative algorithmic challenges. Actively contributed to organizing technical workshops, bootcamps, and intra-university events.
- **General Secretary, Development Wing, IEEE CS CUET Student Branch Chapter** (Present)  
Secretary of the development wing of the IEEE Computer Society student branch chapter at CUET.

- **Joint General Secretary, CUET Debating Society** (2023 – Present)  
Competed in inter-university parliamentary debates, developing high-level logical rebuttals and analytical thinking.  
Helps run the society and organize its inter-departmental debate tournaments.

## REFERENCES

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- Available upon request.